

Timbre Branch Architecture V2

Interface preserving accuracy and research quality revision

Music Genre Classification Research Project
MTG Jamendo 7324 track subsample
Version 2 | September 2026

Decision V2 preserves the complete V1 public tensor contract while revising target preprocessing and branch validation. The default V2 model retains the single 35-output head and accepts the existing full architecture's loss, optimizer, scheduler, and checkpoint policy.

The purpose of V2 is to improve descriptor fidelity through a leakage-safe log-flatness target while remaining directly integrable with the existing shared encoder, joint trainer, other branches, fusion dimensions, and genre head.

Version 2 invariant	Value
Branch input	h_audio FloatTensor [B,128]
Branch output	z_timbre FloatTensor [B,35]
Default neural head	V1 single output head retained
Fusion projection	Fusion owned Linear(35,64)
Primary target change	Training-fitted log transform for spectral_flatness_mean
Primary loss change	Masked Smooth L1 supplied by full trainer

1 Version transition

V2 is an interface-preserving revision. It does not change how the timbre branch connects to the shared encoder, joint trainer, or fusion. It changes the meaning of one standardized target coordinate while leaving loss weighting, optimization, scheduling, and checkpoint selection under full-architecture control.

Design element	V1	V2	Compatibility
Input	[B,128]	[B,128]	Unchanged
Output	[B,35]	[B,35]	Unchanged
Default output head	Single Linear(64,35)	Single Linear(64,35)	Unchanged
Fusion	Linear(35,64)	Linear(35,64)	Unchanged
Epoch budget	Up to 50	Up to 30 under full trainer	Training only
Learning rates	Single optimizer rate	Full-architecture optimizer	Training only
Loss	Cell-averaged masked Smooth L1	Full-trainer masked Smooth L1	Shape safe
Flatness target	Raw then z-score	Log then z-score	New preprocessing state
Ablations	Required in principle	Recommended future evaluation	Evaluation only

1.1 Public contract

Input:	h_audio	FloatTensor [B,128]
Output:	z_timbre	FloatTensor [B,35]
Supervision target:	d_true_standardized	FloatTensor [B,35]
Supervision mask:	valid_mask	BoolTensor [B,35]
Fusion projection:	owned by fusion	Linear(35,64)

Any implementation that changes these shapes, reorders the 35 descriptors, returns a branch-owned fusion token, or exposes h_audio directly to fusion is not Architecture V2.

2 Baseline evidence from results.json

V2 is motivated by the completed V1 joint run rather than by an assumed weakness. The run used 5127 training tracks, 1099 validation tracks, and 1098 held-out test tracks. The best checkpoint occurred at epoch 20, which was also the final recorded epoch.

Baseline measurement	Observed result	V2 interpretation
Best epoch	20 of 20	Training ended while the selection metric was still improving
Validation genre macro AP	0.766	Strong baseline and primary checkpoint-selection signal
Test genre macro AP	0.736	Reasonable held-out generalization
Test genre macro F1	0.622 at threshold 0.5	Useful baseline; threshold tuning is separate
Test genre micro F1	0.636 at threshold 0.5	Reference for full-model comparisons
Test timbre standardized MAE	0.318	Best continuous branch error in this run
Test timbre standardized RMSE	0.561	Some concepts retain substantial residual error
Test timbre macro R squared	0.701	Most declared timbre variance is recoverable
Timbre observed cells	38430	All 1098 test tracks x 35 concepts were evaluated

2.1 Evidence from the training trajectory

Epoch	Validation macro AP	Validation timbre loss	Observation
1	0.358	0.2335	Initial representation
5	0.635	0.1693	Large early improvement
10	0.726	0.1376	Continued learning
15	0.754	0.1174	Smaller improvements
20	0.766	0.1146	Best AP and lowest recorded timbre loss

Because both genre AP and timbre loss improved through the last recorded epoch, V2 adopts the existing full trainer's 30-epoch maximum as the permanent Version 2 budget. Any future increase will be recorded as a later V2 training revision.

3 Timbre fidelity diagnosis

Aggregate timbre performance is strong, but per-feature results are uneven. V2 targets the weak flatness coordinate while preserving well-performing concepts and the complete 35-value contract.

Descriptor	Test R squared	Standardized MAE	Finding
spectral_contrast_mean	0.936	0.138	Strongest spectral concept
mfcc_02_mean	0.931	0.149	Very strong envelope estimate
mfcc_01_mean	0.922	0.166	Very strong envelope estimate
spectral_rolloff_mean	0.881	0.226	Strong high-frequency extent estimate
spectral_flatness_mean	0.090	0.466	Primary V2 preprocessing target
mfcc_13_mean	0.475	0.453	Weak higher-order MFCC mean
mfcc_05_mean	0.486	0.441	Weak mid-order MFCC mean
spectral_centroid_std	0.495	0.475	Weak temporal-variation estimate

3.1 Genre context

Genre	Test average precision	Test F1
classical	0.834	0.748
electronic	0.688	0.574
folk	0.693	0.607
hiphop	0.867	0.779
jazz	0.608	0.361
rock	0.724	0.662

The full-model scores do not prove how much timbre caused the genre result. Jazz is the weakest genre by AP and F1, which reinforces the need for timbre-only and no-timbre comparisons rather than attributing any genre result to timbre from the joint run alone.

4 Default Version 2 neural architecture

```

h_audio [B,128]
-> Linear(128,128)
-> LayerNorm(128)
-> GELU
-> Dropout(0.20)
-> Linear(128,64)
-> GELU
-> Linear(64,35)
-> z_timbre [B,35]
```

The V2 primary experiment deliberately retains the V1 single output head. This isolates the effect of the revised flatness preprocessing while accepting the existing full architecture's training policy. Architecture capacity is not changed in the primary comparison.

Component	Output	Version 2 role
128 to 128 projection	[B,128]	Adapt the pooled shared representation
LayerNorm and GELU	[B,128]	Stable nonlinear mapping
Dropout 0.20	[B,128]	Regularization; initially unchanged
128 to 64 projection	[B,64]	Internal timbre state
Single 64 to 35 head	[B,35]	Ordered standardized concepts

4.1 Descriptor order remains fixed

Indices	Descriptor family	Count
0 to 6	Spectral shape	7
7 to 8	Harmonic and noise	2
9 to 21	MFCC means	13
22 to 34	MFCC standard deviations	13

The output names and order remain those declared in `timbre_branch` constants. Only the preprocessing of `spectral_flatness_mean` changes in V2.

5 Version 2 target preprocessing

5.1 Training-only flatness transform

V1 test fidelity for `spectral_flatness_mean` was substantially weaker than for the other descriptors. V2 addresses its positive, skewed scale before ordinary standardization.

```

epsilon_flatness = fitted or declared positive constant
x_flatness_log = log(max(x_flatness_raw, epsilon_flatness))
z_flatness = (x_flatness_log - train_mean_log) / train_scale_log

```

The transform configuration is fitted or fixed without using validation or test values. Inverse explanations first undo z-scoring and then apply `exp` to return to the original flatness scale.

Requirement	Version 2 rule
Fit population	Training split only
Affected concept	<code>spectral_flatness_mean</code> only
Other 34 concepts	Ordinary training-fitted z-score
Stored checkpoint state	Transform name, epsilon, mean, scale, and feature order
Explanation output	Inverse transformed original acoustic units
Validation and test	Use frozen training transformation unchanged

5.2 Preprocessing version identity

```
preprocessing_version = "timbre_v2_log_flatness_zscore_v1"
```

A V1 checkpoint must not be loaded as V2 without an explicit migration. The tensors have equal shape, but the flatness coordinate uses a different numerical representation.

6 Full architecture loss compatibility

```
prediction = z_timbre [B,35]
target     = d_true_standardized [B,35]
valid_mask = observed_timbre_target_cells [B,35]
L_timbre   = full_trainer_masked_smooth_l1(prediction, target, valid_mask)
```

V2 deliberately accepts the existing full trainer's cell-averaged masked Smooth L1 loss. Missing concepts remain excluded through the published validity mask. Loss weighting is not owned or overridden by the timbre branch, which avoids requiring changes to fusion or joint-training code.

7 Version 2 integration and training protocol

Setting	Version 2 primary protocol
Maximum epochs	30
Early stopping	Not branch-owned; full trainer policy
Optimizer	AdamW
Shared encoder learning rate	Controlled by full trainer
Timbre branch learning rate	Controlled by full trainer
Scheduler	Full trainer schedule
Scheduler factor	Not specified by timbre branch
Scheduler patience	Not specified by timbre branch
Minimum learning rate	Not specified by timbre branch

7.1 Full architecture ownership

```
optimizer: supplied by full trainer
learning rate: supplied by full trainer
scheduler: supplied by full trainer
```

The timbre branch does not create optimizer groups for the shared encoder, other branches, or fusion. The integrated run uses the optimizer and learning-rate policy selected by the existing full-architecture trainer. Those settings must still be recorded in the run configuration.

7.2 Scheduler and checkpoint selection

```
scheduler = full_architecture_scheduler
scheduler policy is external to the timbre branch
checkpoint selection uses the full trainer's validation metric
```

The integrated run follows the existing full trainer's scheduling and checkpoint-selection behavior. The timbre branch checkpoint state records its architecture and preprocessing versions so predictions retain the correct 35-concept meaning.

8 Training sequence

- Validate the 128-dimensional shared-encoder input and 35-dimensional target order.
- Fit the V2 preprocessing state using training rows only.
- Train the unchanged single-head timbre bottleneck within the full model for up to 30 epochs using the full trainer's loss, optimizer, and scheduler.
- Apply the existing full trainer's validation and checkpoint-selection policy.

- Restore the best validation checkpoint.
- Evaluate the test partition once and save per-concept metrics.

9 Amendment rationale

Each V2 amendment responds to either the observed V1 results or the need to isolate scientific causes. The sequence below is part of the experiment design, not an unordered list of options.

Amendment	Why it is included
1 Keep h_audio [B,128]	Preserves compatibility with the shared encoder and avoids changing every branch integration point.
2 Keep z_timbre [B,35]	Preserves the named concept bottleneck, fusion projection, feature order, and explanation interface.
3 Train up to 30 epochs under the full trainer	The 30-epoch maximum is the permanent V2 budget and already matches the existing full trainer.
4 Accept the full trainer scheduler	Scheduling remains owned by the full architecture, avoiding a timbre-specific requirement that cannot be applied independently in joint training.
5 Accept the full trainer optimizer	Optimizer groups and learning rates affect every branch, so V2 does not override decisions owned by the full-architecture trainer.
6 Use the full trainer timbre loss	The existing cell-averaged masked Smooth L1 loss is accepted to preserve compatibility without editing joint-loss code owned by another component.
7 Log-transform spectral flatness	Flatness had R squared 0.090 and RMSE 0.895, far below most concepts. A recorded positive-value transform tests whether target scale is the obstacle.
8 Retain the single output head first	Keeping model capacity fixed allows performance changes to be attributed to preprocessing and training rather than a simultaneous head redesign.
9 Recommend timbre-only and no-timbre ablations	These comparisons remain scientifically useful but are recommendations for the full-architecture owner rather than V2 branch acceptance requirements.
10 Test grouped heads separately as V2-G	A separate run measures descriptor-family specialization without confounding the primary V2 comparison and requires a new checkpoint.
11 Run shape and gradient tests first	Automated tests prevent an internal improvement from silently breaking [B,128] input, [B,35] output, fusion, masks, or gradient flow.

10 Recommended future evaluation

The following controlled comparisons are recommended when the full-architecture owner supports them. They are not required to integrate or accept the V2 timbre branch.

Run	Active timbre route	Research question
V2 Full	Predicted 35 concepts	Primary complete-model result

Run	Active timbre route	Research question
V2 Timbre only	Predicted timbre concepts only	How much genre signal is present in timbre?
V2 No timbre	Timbre fusion mask disabled	What is timbre's incremental contribution?
V2 Oracle timbre	Extracted target concepts	What is the upper bound with perfect concepts?
V2 Shuffled timbre	Concepts assigned to other tracks	Do correct track-level concepts matter?

10.1 Primary comparisons

Incremental timbre value = $AP(V2 \text{ Full}) - AP(V2 \text{ No timbre})$

Prediction bottleneck gap = $AP(V2 \text{ Oracle timbre}) - AP(V2 \text{ Full})$

Report overall macro and micro average precision and F1, plus per-genre changes. Timbre concept metrics include standardized MAE, RMSE, R squared, and per-descriptor results.

11 Shape and gradient contract tests

- Input rank other than two is rejected.
- Input width other than 128 is rejected.
- A valid [B,128] input produces finite [B,35] output.
- Feature names contain exactly 35 unique entries in the published order.
- Fusion projects [B,35] to [B,64] and applies its mask after projection.
- The adapter rejects fusion_token and h_audio bypass fields.
- Gradients propagate from timbre loss through every V2 timbre parameter.
- A branch-local mock downstream test confirms gradients can propagate through the 35-dimensional concept route; production fusion verification remains integration-owned.
- Checkpoint reload reproduces predictions and preprocessing inversion.

12 Separate grouped-head experiment

Grouped internal heads are not part of the primary V2 comparison. They form a separate experiment, identified as V2-G, after the single-head V2 result is established.

```
shared trunk [B,64]
-> spectral_shape_head      -> 7
-> harmonic_noise_head      -> 2
-> mfcc_mean_head           -> 13
-> mfcc_std_head             -> 13
-> concatenate in fixed order -> z_timbre [B,35]
```

Property	V2-G decision
Public input	[B,128], unchanged
Public output	[B,35], unchanged
Fusion	Linear(35,64), unchanged
Internal parameters	Changed; requires a new checkpoint

Property	V2-G decision
Comparison	Matched split, seed, preprocessing, loss, and optimizer budget
Adoption rule	Promote only if validation and repeated-seed evidence improve

12.1 Why it is separated

Changing target preprocessing and head structure simultaneously would prevent attribution. The primary V2 run changes only the flatness target preprocessing while retaining the existing full trainer and V1 head. V2-G then measures the additional effect of descriptor-family specialization.

13 Explainability requirements

- Always expose standardized predictions and inverse-transformed original units.
- Mark weak-fidelity concepts as uncertain rather than treating every concept equally.
- Run descriptor and group interventions through the same fusion pathway used in training.
- Do not claim causal properties of music from model-internal interventions.
- Retain the strict prohibition on direct `h_audio` access by fusion.

14 Version 2 reproducibility package

Artifact	Required content
Architecture report	This V2 specification and version identifier
Source revision	Git commit and tag for all branch and fusion code
Target schema	Exact 35 names and ordering
Preprocessor	Log-flatness settings plus all training means and scales
Data split	TRACK_ID to train, validation, or test assignment
Run configuration	Full-trainer learning rate, scheduler, epochs, seed, and loss weights
Checkpoint	Model, optimizer, preprocessing, and architecture metadata
Results	Per-epoch logs, per-concept metrics, genre metrics, and ablations

14.1 Version identity rules

Change	Version treatment
Full-trainer epoch, scheduler, or learning-rate changes	Full-architecture training revision
Grouped internal heads with the same public contract	V2-G experiment and new checkpoint
Changing the target transformation again	New preprocessing revision
Adding, deleting, or reordering concepts	New major architecture version
Changing the input to temporal encoder tensors	New major architecture version
Adding a direct audio shortcut to fusion	Invalid strict concept-bottleneck model

15 Version 2 acceptance statement

Timbre Branch Architecture V2 is an interface-preserving development of V1. The primary model continues to accept `h_audio` [B,128] and return `z_timbre` [B,35] through the original single output head. V2 adds training-only log preprocessing for spectral flatness, versioned inverse transformation, and branch-local shape, gradient, and checkpoint verification. It accepts the existing full architecture's masked loss, optimizer, scheduler, 30-epoch budget, checkpoint policy, and fusion evaluation workflow.

V2-G remains a separately reported internal-head experiment. Neither V2 nor V2-G may change the public input, output, feature order, or fusion ownership without declaration of a new major version.

16 Completed V2 full architecture results

The completed 30-epoch joint run selected epoch 28 using validation genre macro average precision. The run used 5,127 training tracks, 1,099 validation tracks, and 1,098 held-out test tracks. The test set was evaluated after restoring the selected checkpoint.

Result	V2 value
Training budget	30 epochs
Selected checkpoint	Epoch 28
Validation genre macro AP	0.8129
Test genre macro AP	0.7745
Test genre micro AP	0.7732
Test genre macro F1	0.7036
Test genre micro F1	0.7035
Test binary accuracy	0.8892

16.1 Branch results

Branch	Primary test results	Interpretation
Instrument	macro AP 0.2698; macro F1 0.1138	Improved detection, but rare-label F1 remains limited
Rhythm	MAE 0.4517; RMSE 0.7063; R squared 0.4670	Lower fidelity than the 30-epoch V1 reference
Timbre	MAE 0.4183; RMSE 0.5904; R squared 0.6597	Selected spectral concepts improved, but aggregate fidelity declined
Harmony	MAE 0.3908; RMSE 0.5891; R squared 0.6240	Largest continuous-branch improvement

16.2 Comparison with the 30 epoch V1 reference

Test metric	V1 30 epoch	V2 30 epoch	Change
Genre macro AP	0.7424	0.7745	+0.0321
Genre macro F1	0.6432	0.7036	+0.0604
Instrument macro AP	0.2472	0.2698	+0.0226

Test metric	V1 30 epoch	V2 30 epoch	Change
Rhythm macro R squared	0.5806	0.4670	-0.1136
Timbre macro R squared	0.7043	0.6597	-0.0446
Harmony macro R squared	0.4818	0.6240	+0.1422

V2 improves final genre classification, instrument macro AP, and harmony fidelity. It does not improve aggregate rhythm or timbre R squared. The result therefore supports V2 as a stronger genre model, but not as a uniformly stronger concept predictor. Integration-adaptor experiment I1 must use the complete V1 architecture, including Timbre V1, so that adaptor effects are not confounded with V2 preprocessing.

16.3 Timbre descriptor interpretation

Descriptor	V2 test R squared	Finding
spectral_bandwidth_mean	0.9473	Very strong spectral extent estimate
spectral_rolloff_mean	0.9388	Very strong high-frequency estimate
spectral_centroid_mean	0.9367	Very strong brightness estimate
spectral_contrast_mean	0.8948	Strong spectral contrast fidelity
hnr_mean_db	0.8014	Strong harmonic-to-noise fidelity
inharmonicities_mean	0.6643	Moderate residual error remains
spectral_flatness_mean	0.0654	Weak transformed target; inverse-unit evaluation is required

Because V2 applies a nonlinear log transform to spectral flatness, its standardized flatness errors are not directly equivalent to V1 standardized errors. Future reports must also calculate inverse-transformed flatness error in the original acoustic units. The grouped MFCC-standard-deviation family should be reported explicitly because it is the principal source of aggregate timbre regression.